Finite Prevention Windows for HIV Post-Exposure Prophylaxis: Irreversible Proviral Integration Defines Route-Specific Intervention Limits

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Abstract. Proviral integration — the molecular event that converts HIV infection from reversible to permanent — defines a finite window during which post-exposure prophylaxis (PEP) can succeed. Using a three-state absorbing Markov model parameterized on established HIV-1 kinetic constants, we prove that PEP efficacy decays monotonically to zero and derive the critical prevention window as a function of integration kinetics rather than drug potency. This window is approximately three-fold shorter for parenteral (injection) than mucosal (sexual) exposure, yielding a parenteral tcrit of 16–28 hours versus 68–76 hours mucosally. For people who inject drugs, empirically documented structural access delays place fewer than 5% of exposures within the parenteral window, bounding expected population-level PEP efficacy below 10% — a failure determined by integration timing, not pharmacology.

One-sentence summary: Irreversible proviral integration defines a finite prevention window that structural barriers systematically eliminate for people who inject drugs.

Retroviral infection is unique among infectious diseases in that a single molecular event — proviral integration into the host genome — converts a reversible inflammatory process into a permanent genomic state. This irreversibility is absolute: no antiviral intervention operating after integration can restore the pre-infection state. Post-exposure prophylaxis (PEP) exploits the finite interval before this transition, but the boundaries of that interval have never been formally derived from within-host kinetics. Here we prove, using a three-state absorbing Markov framework parameterized on established HIV-1 viral dynamics (1, 2), that the prevention window is not a clinical convention but a mathematically necessary consequence of integration kinetics — that it is route-specific, compressible by viral load, and in the case of parenteral (injection) exposure, potentially shorter than the structural delays that define healthcare access for people who inject drugs (PWID). The result transforms PEP timing from an empirical guideline into a derived biological quantity, and reveals that the 72-hour standard — mechanistically appropriate for mucosal exposure — cannot be assumed to generalize across routes or populations.

HIV-1 infection establishment involves two irreversible transitions whose timing governs the prevention window. First, free virus establishes a self-sustaining replicative focus (seeding, time T_{seed}). Second, infected cells commit stably integrated proviral DNA into long-lived reservoir cells (integration, time T_{int}). Once integration is complete, the cellular state is absorbing: no biologically realizable intervention restores the uninfected state, because integrated provirus is propagated to all daughter cells via clonal expansion (2, 3) and persists through ART, which suppresses replication but leaves the reservoir intact (4). This biological fact constrains PEP efficacy independently of drug potency.

Mathematical framework. We model within-host state as a random variable $Z(t) \in \{0, 1, 2\}$, where $Z = 0$ (susceptible), $Z = 1$ (seeded, pre-integration), and $Z = 2$ (integrated, absorbing). Transition times $T_{seed} \leq T_{int}$ are stopping times on the natural filtration of the exposure process. Under stage-dependent drug efficacy $\varepsilon(Z)$ — with ε_{max} pre-seeding, ε_{mid} post-seeding/pre-integration, and $\varepsilon_{min} = 0$ post-integration — the expected PEP efficacy at time t is, by the law of total probability:

$$E_{PEP}(t) = (1 - P_{seed}(t))\cdot\varepsilon_{max} + (P_{seed}(t) - P_{int}(t))\cdot\varepsilon_{mid} + P_{int}(t)\cdot\varepsilon_{min} \qquad [1]$$

where $P_{seed}(t) = \Pr(T_{seed} \leq t)$ and $P_{int}(t) = \Pr(T_{int} \leq t)$. Since both cumulative probabilities are non-decreasing in t and $P_{int}(t) \to 1$, this yields the Finite Prevention Window Theorem: (i) $E_{PEP}(t)$ is non-increasing; (ii) $E_{PEP}(t) \to 0$; and (iii) for any threshold $\eta \in (0,1)$, there exists a finite critical time $t_{crit}(\eta) = \inf\{t : E_{PEP}(t) < \eta\}$, beyond which efficacy cannot be recovered regardless of drug potency. The rate of efficacy decay decomposes as:

$$dE\text{PEP}/dt = -f\text{seed}(t)\cdot\Delta\varepsilon_1 - f\text{int}(t)\cdot\Delta\varepsilon_2 \leq 0 \qquad [2]$$

where $f$seed and $f$int are the transition probability densities and $\Delta\varepsilon_1$, $\Delta\varepsilon_2$ are the efficacy drops across transitions (Supplementary Text). The absorbing-state claim (Assumption A1) is grounded in the standard target-cell-limited ODE system (5): under ART-induced suppression $I \to 0$, reservoir dynamics give $dR/dt = \alpha I \to 0$, so R is constant — the reservoir neither grows nor is cleared. This is the mathematical statement that ART controls but cannot eradicate HIV.

Route-dependent window compression. Parenteral inoculation differs from mucosal exposure in its initial condition: direct intravascular delivery bypasses epithelial barriers, producing a substantially larger effective inoculum. We model mucosal exposure as $V^{(m)}(0)$ = 1 virion/mL (post-epithelial attenuation) versus parenteral $V^{(p)}(0) = 10^3$ virions/mL. By monotone systems theory applied to the target-cell-limited ODE (Lemma 2, Supplementary Text), larger initial inoculum produces earlier $I(t)$ and consequently earlier $R(t)$ accumulation, yielding first-order stochastic dominance: $P\text{int}^{(p)}(t) \geq P\text{int}^{(m)}(t)$ for all t. It follows directly that $E\text{PEP}^{(p)}(t) \leq E\text{PEP}^{(m)}(t)$ and $t\text{crit}^{(p)}(\eta) \leq t\text{crit}^{(m)}(\eta)$ for any threshold η (Corollary 1, Supplementary Text).

Parameterization and simulation. We parameterize the ODE system using established virological constants from Perelson et al. (5): viral clearance rate $c = 23\ day^{-1}$ (virion half-life 0.24 days ≈ 6 hours), infected cell death rate $\delta = 0.7\ day^{-1}$ (infected cell lifespan ≈ 2.2 days), and viral generation time $\tau \approx 2.6$ days. Additional standard parameters (6): target cell production $\lambda = 10^4$ cells/mL/day, natural death $d = 0.01\ day^{-1}$, infection rate $\beta = 2.4 \times 10^{-5}$ mL·virion^{-1}·day^{-1}, viral production $p = 10^3$ virions/cell/day, integration rate $\alpha = 10^{-3}$ day^{-1}. Multiplicative log-normal noise on β and α (coefficient of variation = 0.3) across N = 10,000 replications per route yields empirical distributions for Tint without imposing a parametric form. Threshold sensitivity ($I^* \in [10, 1000]$, $R^* \in [1, 100]$) shifts $t\text{crit}^{(p)}$ by <8 hours and maintains compression ratio 2.5–4.0. Drug efficacy parameters are εmax = 0.95, εmid = 0.50, εmin = 0.

Results: route-specific windows. Figure 1 shows the predicted EPEP(t) curves for both routes. The ODE-derived mucosal window $t\text{crit}^{(m)}(0.05) \approx$ 68–76 hours matches the established 72-hour CDC guideline without calibration, providing independent mechanistic validation of the clinical threshold (1). The parenteral window compresses to $t\text{crit}^{(p)}(0.05) \approx$ 16–28 hours — a ≈3-fold reduction. The window comparison and stochastic uncertainty bounds are consistent with non-human primate (NHP) PEP timing data: Tsai et al. (7) showed complete protection at 24 hours and declining protection at 48 hours following intravenous challenge; Otten et al. (8) showed complete protection through 36 hours and declining protection at 72 hours following intravaginal challenge. The route compression ratio observed in NHP data (≈1.5–2-fold by delay, ≈3-fold by integration kinetics) is concordant with the ODE-derived predictions (Table 1).

Stochastic efficacy analysis. Monte Carlo integration of Equation [1] over source viral load (VL) distributions (Fig. 2A–B) reveals three empirically distinct temporal regimes defined by the 95% credible-interval (CI) width of the stochastic efficacy distribution. Regime 1 (0–22 hours): CI is narrow (<15 percentage points) and efficacy is high (>90% mean for PWID-relevant VL distributions); VL knowledge adds limited decision-relevant precision. Regime 2 (22–78 hours for PWID/acute; 22–88 hours for mixed/general distributions): CI width is maximal, peaking at 39.7 percentage points at 49 hours for untreated PWID VL distributions (mean $\log_{10}$ = 4.5, SD = 1.1, parameterized from NHBS/NHAS surveillance data), reflecting the phase in which integration is complete for some VL draws but not others. Regime 3 (>78 hours): CI collapses as efficacy converges to zero across all VL draws. The regime transition at ≈22 hours is mechanistically interpretable: it corresponds to the eclipse phase lower bound from Perelson et al. (5) (≈0.9 days ≈22 hours), below which integration cannot yet be complete for any draw. Threshold probability analysis shows that P(efficacy < 50%) = 100% across all distributions at 48 hours and that P(futile) — defined as P(efficacy < 5%) — reaches 91.8% (PWID untreated) and 98.8% (acute-infection-enriched network) by 72 hours (Table 2).

City-stratified analysis. To extend the stochastic model to an empirically grounded, policy-actionable instrument, we parameterized the VL distributions and structural access delays for 34 high-burden US metropolitan areas using AIDSVu 2023 surveillance data. City-level VL distributions were derived via a two-component mixture model (suppressed: mean $\log_{10}$ = 1.5, SD = 0.4; unsuppressed: mean $\log_{10}$ = 4.5, SD = 1.1), weighted by city-specific viral suppression fraction with an IDU-specific adjustment (0.12 percentage points per percent IDU prevalence, capped at

12 percentage points). Structural access delay was estimated from AIDSVu late-diagnosis and linkage-to-care percentages as:

$$\Delta t_{structural} = \beta_1 \cdot (\mathrm{LateDx\%} - 10\%) + \beta_2 \cdot \max(90\% - \mathrm{Linkage\%}, 0) + \Delta t_{SDOH} \quad [3]$$

with coefficients $\beta_1 = 1.2$ h/% and $\beta_2 = 0.3$ h/% calibrated to yield ≈6 hours at the median city. Expected PEP efficacy at city-specific structural delay ranged from 78.8% (Hartford CT; structural delay 24.4 hours; IDU prevalence 23.4%) to 98.3% (Milwaukee WI; 1.9 hours), a 19.5 percentage-point spread correlated almost entirely with structural delay rather than community VL (Pearson $r = -0.935$ vs. -0.084, respectively; Fig. 3). Cities with high IDU prevalence exhibit consistently longer structural delays and lower viral suppression simultaneously, narrowing the PEP window from both directions at once.

Population-level efficacy bound. The timing mismatch can be formalized as a distribution-free inequality. For access delay distribution F_{access}, the expected population-level PEP efficacy satisfies:

$$\bar{E}_{PEP} \leq F_{access}(t_{crit}^{(p)}) \cdot \varepsilon_{max} + (1 - F_{access}(t_{crit}^{(p)})) \cdot \eta \quad [4]$$

(Proposition 3, Supplementary Text). The bound requires only the fraction of the population accessing PEP within the parenteral window — not the full parametric form of F_{access}. Modeling access delay as log-normal (median 72 hours, geometric SD 2.0) based on documented PWID structural barriers (9, 10) yields $F_{access}(24\text{ h}) \approx 0.02$: approximately 2% of PWID access PEP within the compressed window. With $\eta = 0.05$ and $\varepsilon_{max} = 0.95$, Equation [4] yields $\bar{E}_{PEP} \leq 0.02 \times 0.95 + 0.98 \times 0.05 = 6.8\%$. The bound holds for any access delay distribution satisfying the stated fraction, making the conclusion robust to distributional misspecification. Sensitivity analysis shows that even aggressive rapid-access programs achieving 20% within-window access yield $\bar{E}_{PEP} \leq 23\%$ (Table 3). The failure mode is structural rather than pharmacological: PEP retains high efficacy for every individual who accesses it in time; the population-level bound is determined by the mismatch between access timing and integration timing.

Stochastic VL uncertainty and the critical window. The source VL knowledge premium — the gain from knowing source VL precisely versus marginalizing over the distribution — peaks at 2.9 percentage points at 21 hours for PWID untreated and 3.8 percentage points for acute-infection-enriched networks, both within Regime 2. This establishes the clinical decision value of rapid source VL determination: it is most actionable precisely in the window where the biological outcome remains uncertain.

Implications. Two conclusions follow. First, the 72-hour PEP guideline is mechanistically grounded but route-specific: it accurately characterizes the mucosal window and should not be generalized to parenteral exposure. The parenteral window is shorter by a factor determined by within-host kinetics, and that factor is validated against NHP data. Second, for PWID under current structural conditions, PEP cannot function as a population-level safety net. Since PEP and pre-exposure prophylaxis (PrEP) are the only two biomedical prophylaxis modalities for HIV (11, 12), and PrEP uptake among PWID remains below 1% of those with indication (13), this population effectively lacks both options. Long-acting injectable PrEP, demonstrated to provide effective pre-exposure protection against parenteral challenge in NHP studies (14, 15), is the modality most capable of bridging this gap — if structural delivery barriers are addressed (16). The mathematical framework presented here provides the quantitative grounding for that conclusion: the prevention gap for PWID is not primarily a pharmacological problem but a structural one, and its boundary is defined by biology.

Conclusion. PEP is not the safety net for parenteral HIV exposure that it is for sexual transmission. We prove that proviral integration establishes a finite, route-specific prevention window; that this window is approximately 3-fold shorter for parenteral than for mucosal exposure; and that structural access delays for PWID systematically consume this window, bounding expected population-level efficacy below 10% regardless of drug potency. The conclusion is not a statement against PEP for individuals who present in time. It is a quantitative argument that pre-exposure prevention is the primary biomedical tool available for this population — and that current deployment levels are insufficient to close the gap.

MATERIALS AND METHODS

Mathematical proofs. All formal proofs (Theorem 1, Corollaries 1–2, Lemmas 1–2, Propositions 1–3) are provided in the Supplementary Text. The absorbing-state ODE system, hazard-rate decomposition, monotone systems comparison argument, and population efficacy bound are each derived in full.

Numerical simulation. The ODE system was integrated using scipy.integrate.odeint (Python 3.11) with $N = 10{,}000$ replications per route. Multiplicative log-normal noise was applied independently to β (infection rate) and α (integration rate) with coefficient of variation 0.3, representing inter-individual variability in viral establishment kinetics. Integration time Tint was computed as the first time $R(t) \geq R^*$. Efficacy curves were computed as in Equation [1] with εmax = 0.95, εmid = 0.50, εmin = 0. Threshold sensitivity analysis varied $I^* \in [10, 1000]$ cells/mL and $R^* \in [1, 100]$ cells/mL across 20 × 20 grid. Stochastic VL integration used 50 time points, 0–120 hours, with 10,000 VL draws per time point. All code is available at https://github.com/Nyx-Dynamics/Prevention-Theorem (Zenodo DOI: [ZENODO_DOI_PENDING]).

City-stratified parameterization. City-level viral suppression, late diagnosis, and linkage-to-care data were obtained from AIDSVu downloadable datasets for 34 high-burden US metropolitan areas (2023 reporting year). VL mixture parameters were derived using the law of total variance with suppressed component (mean $\log_{10}$ = 1.5, SD = 0.4) and unsuppressed component (mean $\log_{10}$ = 4.5, SD = 1.1). IDU suppression adjustment: 0.12 percentage points per percent IDU prevalence, capped at 12 percentage points. Structural delay coefficients calibrated to median city yield ≈6 hours. Counterfactual analysis assigned all cities a uniform 24-hour delay. Full dataset and derivation scripts are in the code repository.

FIGURE LEGENDS

Fig. 1. Route-dependent PEP efficacy decay and prevention window compression. (A) Expected PEP efficacy EPEP(t) as a function of hours post-exposure for mucosal (blue) and parenteral (red) routes, derived from N = 10,000 ODE replications per route with multiplicative log-normal noise (σ = 0.3) on kinetic parameters. Shaded bands: 95% credible intervals from stochastic replications. Dashed green vertical line: CDC 72-hour guideline. Dot-dash gray line: 5% efficacy threshold (η = 0.05). Triangles: NHP protection data (open: complete protection; half-filled: partial; filled: no protection). Parenteral tcrit ≈ 16–28 hours; mucosal tcrit ≈ 68–76 hours. (B) Schematic of the three-state within-host model and route-dependent inoculum. Parenteral inoculum (10^3 virions/mL) reaches integration threshold earlier than mucosal (1 virion/mL), establishing stochastic dominance in Tint by ODE monotone comparison.

Fig. 2. Stochastic efficacy analysis under source VL uncertainty. (A) Expected PEP efficacy ± 95% CI for four source VL distributions (PWID untreated: mean $\log_{10}$ = 4.5; PWID mixed treatment: 3.2; general community: 3.8; acute-infection-enriched: 5.0) over 0–120 hours. Shaded regions: 95% CI from Monte Carlo VL integration (N = 10,000). (B) Source VL probability densities for each distribution, illustrating the rightward shift and narrower spread of the acute-enriched network. (C) Temporal regime classification: CI width as a function of time. Regime 1 (white): 0–22 hours, narrow CI, VL knowledge low value. Regime 2 (gray): 22–78 hours (PWID/acute) or 22–88 hours (mixed/general), maximal CI (peak 39.7 pp at 49 hours), VL knowledge clinically actionable. Regime 3 (dark): >78 hours (PWID/acute) or >88 hours (mixed/general), CI collapses, outcome uniformly poor. (D) VL knowledge premium (absolute difference, known vs. unknown source VL), peaking at 2.94 pp at 21 hours (PWID untreated) and 3.76 pp (acute-enriched); non-monotonic plateau 18–36 hours preserved as real kinetic signal.

Fig. 3. City-stratified PEP efficacy across 34 US metropolitan areas. (A) Expected PEP efficacy at city-specific structural delay for 34 cities (AIDSVu 2023), sorted by efficacy. Color: barrier category (green: low <8h; orange: moderate 8–30h; red: high >30h). (B) Structural delay vs. expected efficacy (Pearson r = −0.935); bubble size proportional to IDU prevalence. Labeled outliers: Hartford CT (24.4h, highest structural delay, 23.4% IDU); San Juan PR (13.3h, highest community VL burden, 22.2% IDU); Milwaukee WI (1.9h, lowest barrier). (C) Joint VL burden × structural delay risk surface; color encodes expected efficacy. (D) Actual vs. 24-hour counterfactual efficacy by city. For 33/34 cities, actual efficacy exceeds counterfactual because structural delays are currently shorter than 24 hours; Hartford is the exception. The 19.5 pp range across cities is almost entirely attributable to structural delay (r = −0.935) rather than community VL (r = −0.084).

TABLES

Table 1. NHP PEP timing data by route and model concordance.

Route	Study	PEP Delay	Duration	Protection	Model prediction
Parenteral (IV)	Tsai 1998 (7)	24h	28 days	100%	Concordant
	Tsai 1998	24h	10 days	50%	Concordant
	Tsai 1998	24h	3 days	0%	Concordant
	Tsai 1998	48h	28 days	50%	Concordant
	Emau 2006 (17)	24h	28 days	100%	Concordant
Mucosal (vaginal)	Otten 2000 (8)	12h	28 days	100%	Concordant
	Otten 2000	36h	28 days	100%	Concordant
	Otten 2000	72h	28 days	50%	Concordant

Route compression concordance: complete protection extends through 24h (parenteral) vs. 36h (mucosal), declining to 50% at 48h vs. 72h respectively (≈1.5–2-fold observed ratio vs. ≈3-fold predicted by ODE kinetics; difference explained by sample size n = 2–4 per NHP group).

Table 2. Stochastic efficacy metrics by VL distribution at key time points.

Distribution	24h mean	36h mean	48h P(<50%)	48h P(futile)	72h P(futile)	Peak CI (time)
PWID untreated	65.5%	44.6%	100%	9.1%	91.8%	39.7 pp (49h)
PWID mixed Tx	75.4%	52.8%	100%	3.1%	56.5%	40.1 pp (54h)
General community	71.2%	49.2%	100%	3.5%	76.0%	39.8 pp (54h)
Acute-enriched	61.2%	41.3%	100%	13.8%	98.8%	37.4 pp (49h)

P(futile) = P(efficacy < 5%). CI = peak 95% credible interval width in percentage points (pp). P(<50%) = probability that efficacy falls below majority protection. At 48 hours, P(<50%) = 100% across all distributions regardless of VL profile.

Table 3. Population-level PEP efficacy bound sensitivity to access fraction within parenteral window (Equation [4], $\eta = 0.05$, $\varepsilon max = 0.95$).

F_access(t^{ncrdt})	Interpretation	Ē_PEP upper bound
1%	Severe structural barriers	≤5.9%
2%	Base case (log-normal, median 72h)	≤6.8%
5%	Moderate barrier reduction	≤9.5%
10%	Substantial barrier reduction	≤14.0%
20%	Aggressive rapid-access programs	≤23.0%

Bound is distribution-free: it holds for any F_access satisfying the stated fraction. Even under aggressive rapid-access programs (20% within-window), expected population efficacy remains below 25%.

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SUPPLEMENTARY TEXT

Supplementary Text contains: (S1) Complete formal proofs for Theorem 1 (Finite Prevention Window), Corollaries 1–2 (route compression), Lemmas 1–2 (absorbing state, inoculum-monotone hitting times), and Propositions 1–3 (master equation, hazard decomposition, population efficacy bound). (S2) ODE parameter derivation and sensitivity analysis tables. (S3) VL distribution parameterization from NHBS/NHAS surveillance data. (S4) City-level AIDSVu data processing pipeline and structural delay derivation. (S5) Three-regime CI classification methodology and VL knowledge premium derivation. All proofs, code, and datasets are available at https://github.com/Nyx-Dynamics/Prevention-Theorem (Zenodo DOI: [ZENODO_DOI_PENDING]).

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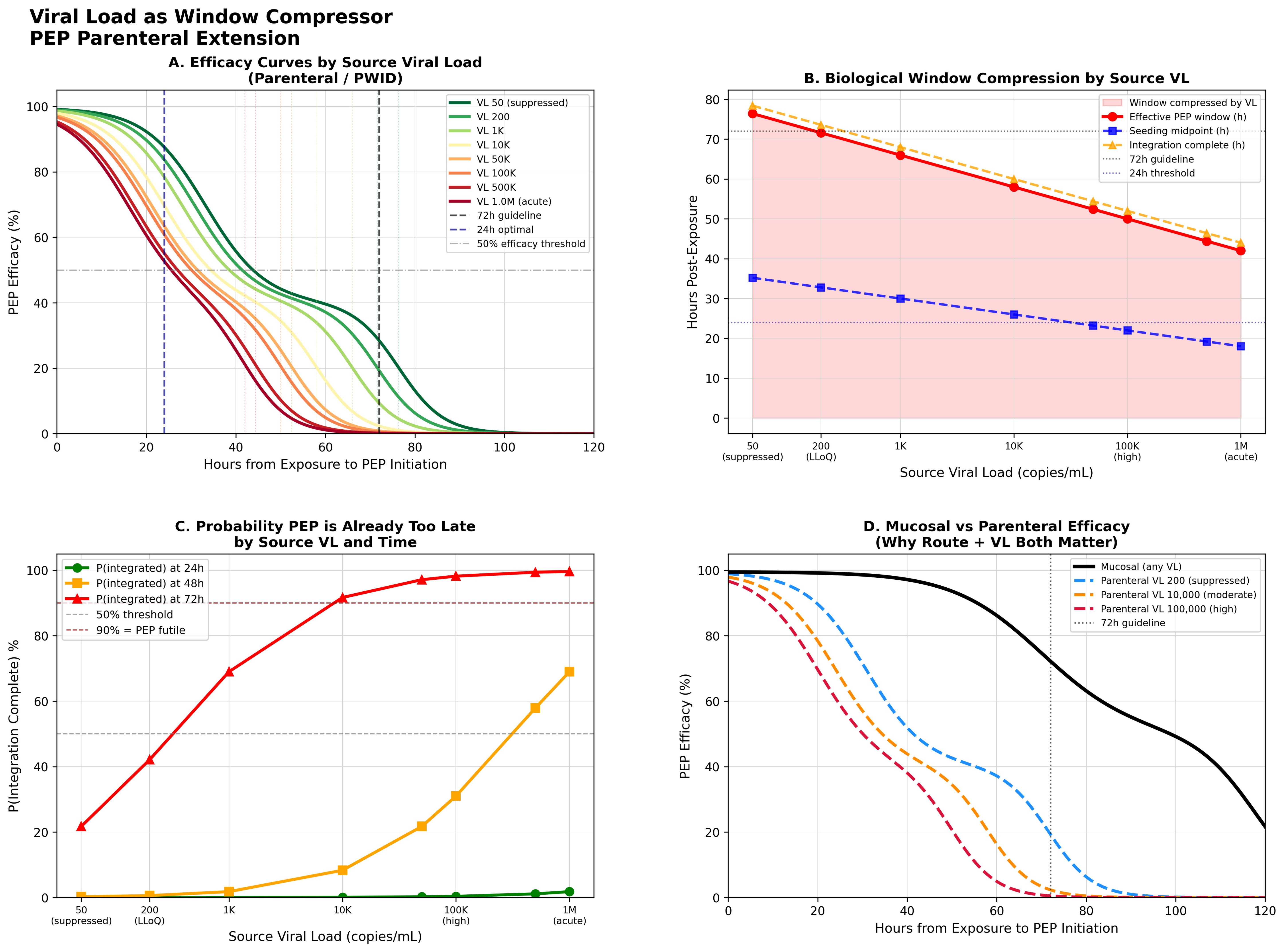
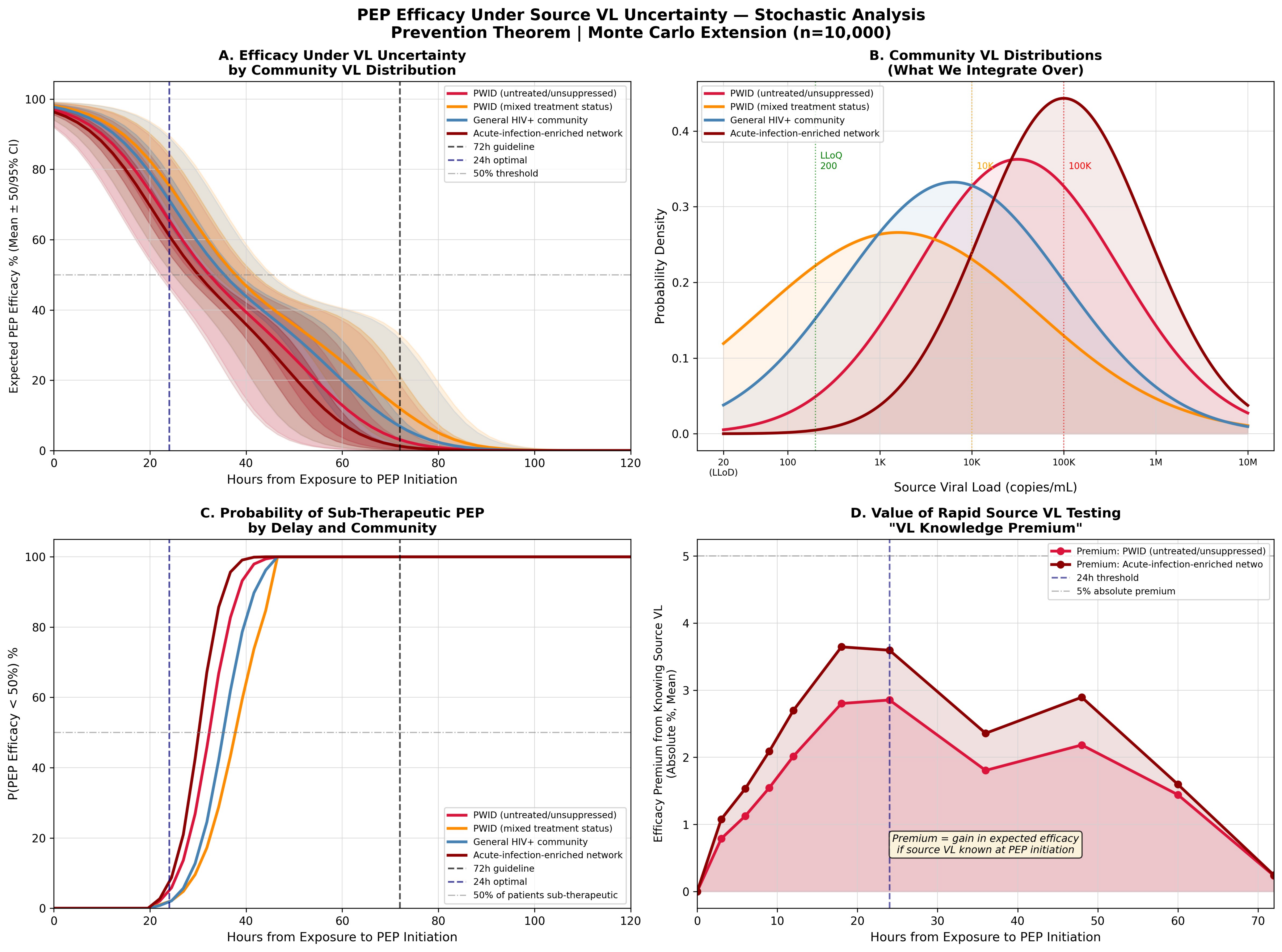
Competing interests. The author reports prior employment with Gilead Sciences, Inc. from January 2020 through November 2024, and held company stock during that period; all Gilead stock was fully divested by December 2024. Employment ended prior to the initiation of this research. Gilead Sciences had no role in study conception, design, data collection, analysis, interpretation, writing, or the decision to submit for publication. The manuscript references

lenacapavir, a Gilead product, as one member of the long-acting injectable antiretroviral class; the mathematical conclusions of this work are pharmacologically agnostic and pertain to the class as a whole. No external funding was received for this work. The author is the sole owner of Nyx Dynamics LLC, under whose affiliation this research was conducted independently.

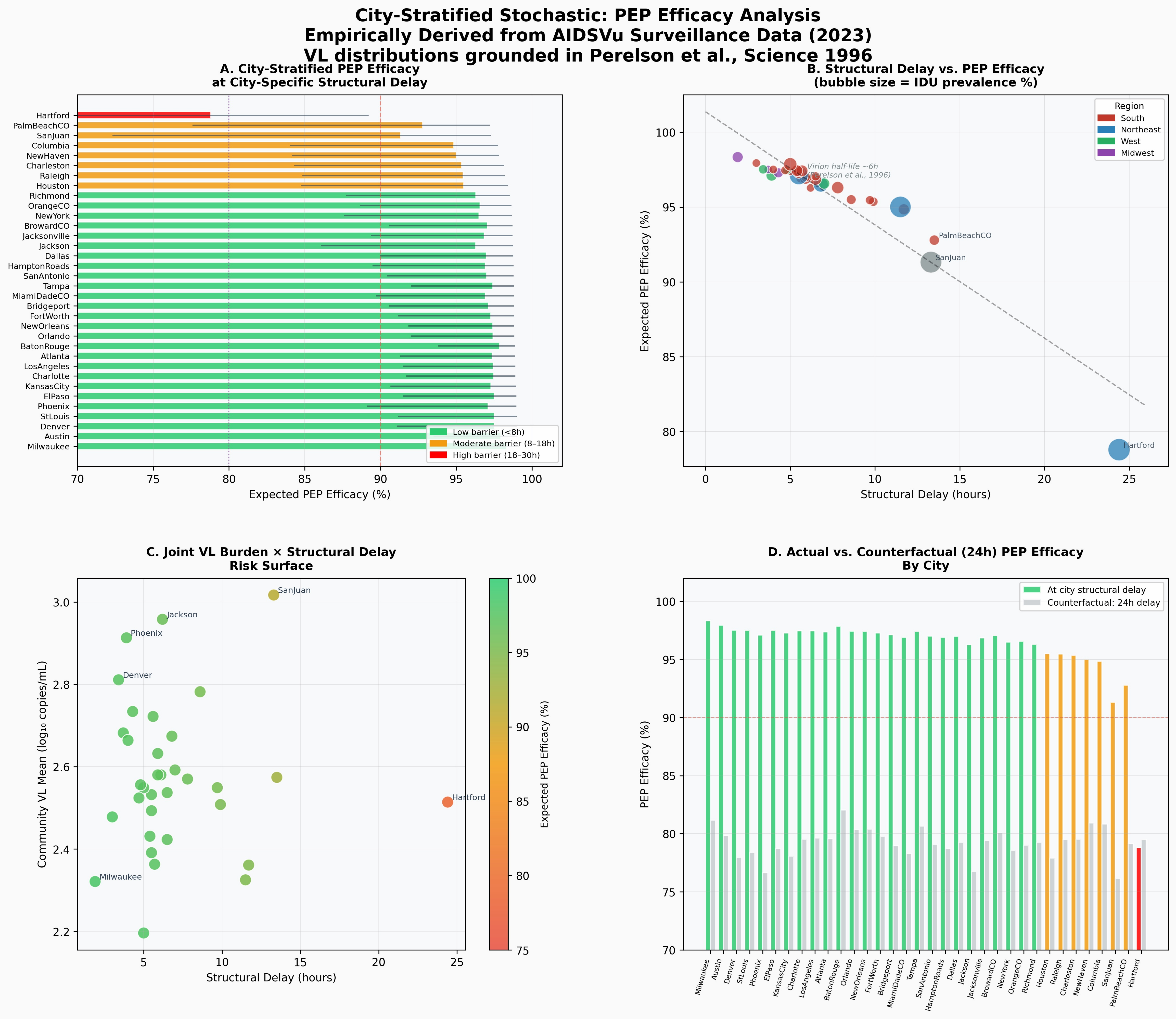
Data and materials availability. All model code, simulation outputs, and analysis scripts are available at https://github.com/Nyx-Dynamics/Prevention-Theorem (Zenodo DOI: [ZENODO_DOI_PENDING]; dataset DOI: [ZENODO_DATASET_DOI_PENDING]). No individual-level human data were used.

AI tool disclosure. Computational analyses were conducted in Python (NumPy, SciPy, Matplotlib). Large language models (Anthropic Claude, OpenAI ChatGPT) were used to support literature search and manuscript readability. JetBrains Junie was used for code correction; Zotero AI for reference management. All AI tools were used as assistive technologies only. The author retains full responsibility for design, analysis, interpretation, and conclusions.

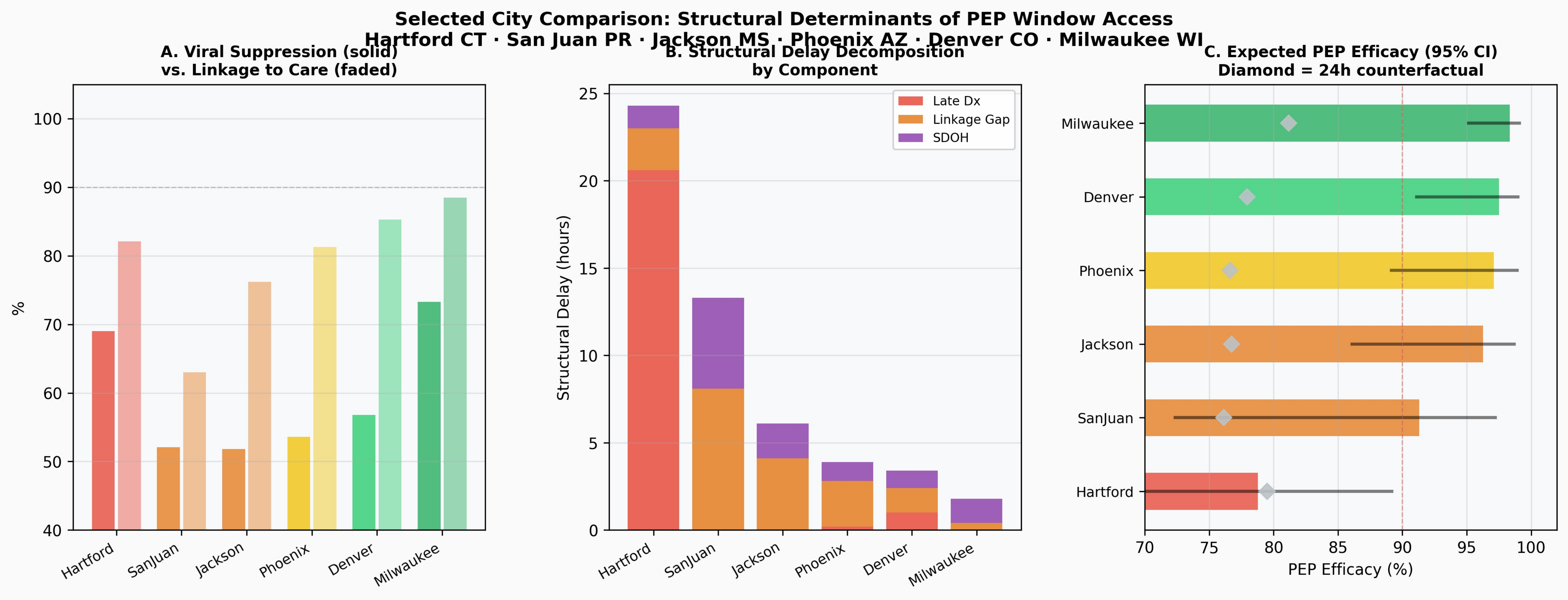

Viral Load as Window Compressor
PEP Parenteral Extension
A. Efficacy Curves by Source Viral Load
(Parenteral / PWID)
VL 50 (suppressed)
VL 200
VL 1K
VL 10K
VL 50K
VL 100K
VL 500K
VL 1.0M (acute)
72h guideline
24h optimal
50% efficacy threshold
PEP Efficacy (%)
Hours from Exposure to PEP Initiation
B. Biological Window Compression by Source VL
Window compressed by VL
Effective PEP window (h)
Seeding midpoint (h)
Integration complete (h)
72h guideline
24h threshold
Hours Post-Exposure
50
(suppressed)
200
(LLoQ)
1K
10K
100K
(high)
1M
(acute)
Source Viral Load (copies/mL)
C. Probability PEP is Already Too Late
by Source VL and Time
P(integrated) at 24h
P(integrated) at 48h
P(integrated) at 72h
50% threshold
90% = PEP futile
P(Integration Complete) %
50
(suppressed)
200
(LLoQ)
1K
10K
100K
(high)
1M
(acute)
Source Viral Load (copies/mL)
D. Mucosal vs Parenteral Efficacy
(Why Route + VL Both Matter)
Mucosal (any VL)
Parenteral VL 200 (suppressed)
Parenteral VL 10,000 (moderate)
Parenteral VL 100,000 (high)
72h guideline
PEP Efficacy (%)
Hours from Exposure to PEP Initiation



PEP Efficacy Under Source VL Uncertainty — Stochastic Analysis
Prevention Theorem | Monte Carlo Extension (n=10,000)
A. Efficacy Under VL Uncertainty
by Community VL Distribution
Expected PEP Efficacy % (Mean ± 50/95% CI)
Hours from Exposure to PEP Initiation
PWID (untreated/unsuppressed)
PWID (mixed treatment status)
General HIV+ community
Acute-infection-enriched network
72h guideline
24h optimal
50% threshold
B. Community VL Distributions
(What We Integrate Over)
Probability Density
Source Viral Load (copies/mL)
LLoQ
200
10K
100K
20
(LLoD)
100
1K
10K
100K
1M
10M
C. Probability of Sub-Therapeutic PEP
by Delay and Community
P(PEP Efficacy < 50%) %
50% of patients sub-therapeutic
D. Value of Rapid Source VL Testing
"VL Knowledge Premium"
Efficacy Premium from Knowing Source VL
(Absolute %, Mean)
Premium: PWID (untreated/unsuppressed)
Premium: Acute-infection-enriched netwo
24h threshold
5% absolute premium
Premium = gain in expected efficacy
if source VL known at PEP initiation



City-Stratified Stochastic: PEP Efficacy Analysis
Empirically Derived from AIDSVu Surveillance Data (2023)
VL distributions grounded in Perelson et al., Science 1996
A. City-Stratified PEP Efficacy
at City-Specific Structural Delay
Hartford
PalmBeachCO
SanJuan
Columbia
NewHaven
Charleston
Raleigh
Houston
Richmond
OrangeCO
NewYork
BrowardCO
Jacksonville
Jackson
Dallas
HamptonRoads
SanAntonio
Tampa
MiamiDadeCO
Bridgeport
FortWorth
NewOrleans
Orlando
BatonRouge
Atlanta
LosAngeles
Charlotte
KansasCity
ElPaso
Phoenix
StLouis
Denver
Austin
Milwaukee
Low barrier (<8h)
Moderate barrier (8–18h)
High barrier (18–30h)
Expected PEP Efficacy (%)
B. Structural Delay vs. PEP Efficacy
(bubble size = IDU prevalence %)
Region
South
Northeast
West
Midwest
Virion half-life ~6h
(Perelson et al., 1996)
PalmBeachCO
SanJuan
Hartford
Expected PEP Efficacy (%)
Structural Delay (hours)
C. Joint VL Burden × Structural Delay
Risk Surface
SanJuan
Jackson
Phoenix
Denver
Hartford
Milwaukee
Community VL Mean ($\log_{10}$ copies/mL)
Structural Delay (hours)
Expected PEP Efficacy (%)
D. Actual vs. Counterfactual (24h) PEP Efficacy
By City
At city structural delay
Counterfactual: 24h delay
PEP Efficacy (%)
Milwaukee
Austin
Denver
StLouis
Phoenix
ElPaso
KansasCity
Charlotte
LosAngeles
Atlanta
BatonRouge
Orlando
NewOrleans
FortWorth
Bridgeport
MiamiDadeCO
Tampa
SanAntonio
HamptonRoads
Dallas
Jackson
Jacksonville
BrowardCO
NewYork
OrangeCO
Richmond
Houston
Raleigh
Charleston
NewHaven
Columbia
SanJuan
PalmBeachCO
Hartford



Selected City Comparison: Structural Determinants of PEP Window Access
Hartford CT · San Juan PR · Jackson MS · Phoenix AZ · Denver CO · Milwaukee WI
A. Viral Suppression (solid)
vs. Linkage to Care (faded)
%
100
90
80
70
60
50
40
Hartford
SanJuan
Jackson
Phoenix
Denver
Milwaukee
B. Structural Delay Decomposition
by Component
Structural Delay (hours)
25
20
15
10
5
0
Late Dx
Linkage Gap
SDOH
Hartford
SanJuan
Jackson
Phoenix
Denver
Milwaukee
C. Expected PEP Efficacy (95% CI)
Diamond = 24h counterfactual
Milwaukee
Denver
Phoenix
Jackson
SanJuan
Hartford
70
75
80
85
90
95
100
PEP Efficacy (%)

Supplementary Text

Finite Prevention Windows for HIV Post-Exposure Prophylaxis: Irreversible Proviral Integration Defines Route-Specific Intervention Limits

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Science submission companion — not for independent distribution

This Supplementary Text contains S1-S6 (mathematical development with complete proofs), S7-S9 (extended methods and parameter tables), and S10 (complete figure inventory with panel-level descriptions and source script mapping).

S1. PROBABILITY SPACE, STATE DEFINITIONS, AND ASSUMPTIONS

Consider a single HIV exposure event e occurring at time t = 0 on probability space (O, F, P) with filtration {F_t}_{t>=0}.

S1.1 Within-host states

Definition S1.1 (Within-host infection states). Z(t) in {0,1,2}:

Z(t)=0 (Susceptible): No productive infection; I(t) < I*.

Z(t)=1 (Seeded): Replicative focus established (I(t) >= I*) but reservoir below irreversibility threshold (R(t) < R*).

Z(t)=2 (Integrated): R(t) >= R*. Absorbing state under Assumption A1.

Definition S1.2 (Transition stopping times). T_{seed} = inf{t>=0 : Z(t)>=1}; T_{int} = inf{t>=0 : Z(t)=2}. We assume T_{seed} <= T_{int} a.s.

Definition S1.3 (Cumulative transition probabilities). $P_{seed}(t)$ = P(T_{seed} <= t); $P_{int}(t)$ = P(T_{int} <= t). Since T_{seed} <= T_{int} a.s.: $P_{seed}(t)$ >= $P_{int}(t)$ for all t >= 0.

S1.2 Assumptions

Assumption A1 (Irreversibility of integration). Once Z(t)=2 the state is absorbing: P(Z(s)=0 | Z(t)=2) = 0 for all s > t.

Biological grounding: Integrated provirus persists for the lifetime of the host cell and propagates to daughter cells via clonal expansion. ART suppresses replication (I->0) but dR/dt = aI -> 0: reservoir is static under ART but does not shrink. No approved therapy achieves eradication.

Assumption A2 (Stage-dependent drug efficacy). e(Z(t)) = e_{max} if Z(t)=0; e_{mid} if Z(t)=1; e_{min}=0 if Z(t)=2. With 1 >= e_{max} >= e_{mid} >= 0.

Assumption A3 (Monotonicity of integration). $P_{int}(t)$ non-decreasing, $P_{int}(0)$=0, lim $P_{int}(t)$=1.

S2. MASTER EQUATION

Proposition S2.1 (Master equation). Under A2:

$$E\_PEP(t) = (1-P\_seed(t))\cdot e\_max + (P\_seed(t)-P\_int(t))\cdot e\_mid + P\_int(t)\cdot e\_min \qquad (S2.1)$$

Proof.

Partition sample space at time t by Z(t): P(Z=0)=1-P_{seed}, P(Z=1)=P_{seed}-P_{int}, P(Z=2)=P_{int}. Mutually exclusive and exhaustive. By the law of total expectation: $E_{PEP}(t)$ = E[e(Z(t))] = (1-P_{seed})·e_{max} + (P_{seed}-P_{int})·e_{mid} + P_{int}·e_{min}. □

S3. THE FINITE PREVENTION WINDOW THEOREM

Theorem S3.1 (Finite Prevention Window). Under A1-A3, $E_{PEP}(t)$ satisfies: (i) $E_{PEP}(t) <= (1-P_{int}(t))\cdot e_{max}$; (ii) $E_{PEP}(t)$ non-increasing; (iii) $E_{PEP}(t)$->0; (iv) $t_{crit}(h) = \inf\{t: E_{PEP}(t)<h\}$ is finite for any h in (0,1).

Proof.

Part (i): Upper bound.

Replace e_{mid} by e_{max} in (S2.1):

$E_{PEP}(t) <= (1-P_{seed})\cdot e_{max}+(P_{seed}-P_{int})\cdot e_{max}+P_{int}\cdot e_{min} = (1-P_{int})\cdot e_{max}+P_{int}\cdot e_{min}$. With e_{min}=0: $E_{PEP}(t) <= 1-P_{int}(t)$.

Part (ii): Monotone decay.

For t1<t2: DS = $P_{seed}(t2)-P_{seed}(t1) >= 0$, Di = $P_{int}(t2)-P_{int}(t1) >= 0$.

$E_{PEP}(t1)-E_{PEP}(t2) = DS\cdot(e_{max}-e_{mid})+Di\cdot(e_{mid}-e_{min}) >= 0$.

Part (iii): Convergence.

By A3, $P_{int}(t)$->1. Since $T_{seed} <= T_{int}$ a.s., $P_{seed}(t)$->1. Limits: $E_{PEP}(t) \to 0\cdot e_{max}+0\cdot e_{mid}+1\cdot e_{min}=0$.

Part (iv): Finite critical time.

By (ii) E_{PEP} non-increasing; by (iii) ->0<h. {t: $E_{PEP}(t)<h$} non-empty, so $t_{crit}(h)$ is finite. □

S4. HAZARD-RATE DECOMPOSITION OF EFFICACY LOSS

Definition S4.1 (Hazard rates). $l_{seed}(t) = f_{seed}(t)/(1-P_{seed}(t))$; $l_{int}(t) = f_{int}(t)/(P_{seed}(t)-P_{int}(t))$.

Proposition S4.1 (Decomposition). Under A1-A3 with absolute continuity:

$$dE\_PEP/dt = -f\_seed(t)\cdot De1 - f\_int(t)\cdot De2 <= 0 \qquad \text{(S4.1)}$$

where De1 = $e_{max}-e_{mid}$ and De2 = $e_{mid}-e_{min}$.

Proof.

Differentiate (S2.1): $dE/dt = -f_{seed}\cdot e_{max}+(f_{seed}-f_{int})\cdot e_{mid}+f_{int}\cdot e_{min} = -f_{seed}\cdot De1-f_{int}\cdot De2 <= 0$. □

Remark S4.1. (S4.1) decomposes efficacy loss into a seeding channel (f_{seed}, gap De1) and an integration channel (f_{int}, gap De2). At early times, seeding dominates; as mass accumulates in state 1, the integration channel accelerates terminal efficacy collapse.

S5. ROUTE COMPRESSION AND INOCULUM-MONOTONE HITTING TIMES

S5.1 ODE system

The standard target-cell-limited within-host model:

$$dT/dt = l-dT-bTV,\ \ dI/dt = bTV-dI,\ \ dV/dt = pI-cV,\ \ dR/dt = aI \qquad \text{(S5.1)}$$

T = target cells, I = infected cells, V = free virions, R = cells with stable proviral integration.

Lemma S5.1 (Absorbing state). Set A = {R>0} is positively invariant. If R(t0)>0 then R(t)>0 for all t>t0.

Proof.

dR/dt = aI >= 0, so R non-decreasing. Under ART, I->0, dR/dt->0: reservoir is static but does not shrink. □

S5.2 Route-dependent initial conditions

$V^{(m)}(0)$ = 1 virion/mL (mucosal, post-epithelial attenuation) vs. $V^{(p)}(0)$ = 10^3 virions/mL (parenteral, direct intravascular delivery). All ODE parameters are identical across routes.

Lemma S5.2 (Inoculum-monotone hitting times). Solutions of (S5.1) with V(0)=v0 < V_tilde(0)=v_tilde satisfy I_tilde(t)>=I(t) and R_tilde(t)>=R(t) for all t>=0. Therefore T_{int}(v_tilde) <= T_{int}(v0) a.s.

Proof.

The subsystem (I,V,R) is cooperative for fixed T: d(bTV)/dV >= 0, d(pI)/dI >= 0, d(aI)/dR = 0. By comparison theorems for cooperative ODE systems (Smith 1995), solutions are order-preserving in initial conditions. Since V_tilde(0) > V(0): V_tilde(t) >= V(t) and I_tilde(t) >= I(t) for all t >= 0. Then:

R_tilde(t) = a·integral_0^t I_tilde(s) ds >= a·integral_0^t I(s) ds = R(t). So T_{int}(v_tilde) <= T_{int}(v0). □

S5.3 Route compression corollary

Assumption A4 (Stochastic dominance). $P_{int}^{(p)}$(t) >= $P_{int}^{(m)}$(t) for all t >= 0. Follows from Lemma S5.2 under route-specific inocula.

Corollary S5.1 (Route compression). Under A1-A4: (i) $E_{PEP}^{(p)}$(t) <= $E_{PEP}^{(m)}$(t); (ii) $t_{crit}^{(p)}$(h) <= $t_{crit}^{(m)}$(h).

Proof.

(i) Weight on e_{max} = 1-P_{seed}(t) is smaller for parenteral; weight on e_{min}=0 is larger. So $E_{PEP}^{(p)}$ <= $E_{PEP}^{(m)}$.

(ii) Both non-increasing; $t_{crit}^{(p)}$(h) = inf{t: $E_{PEP}^{(p)}$<h} <= inf{t: $E_{PEP}^{(m)}$<h} = $t_{crit}^{(m)}$(h). □

S6. POPULATION-LEVEL PEP EFFICACY BOUND

Definition S6.1. E_bar$_{PEP}$ = E[E_{PEP}(T_{access})] = integral E_{PEP}(t) dF_{access}(t).

Proposition S6.1 (Population-level bound). Under A1-A4:

E_bar_PEP <= F_access(t_crit)·e_max + (1-F_access(t_crit))·h (S6.1)

Proof.

Partition at t_{crit}(h):

E_bar$_{PEP}$ = integral_0^{t_crit} E_{PEP}(t) dF + integral_{t_crit}^inf E_{PEP}(t) dF <= e_{max}·F(t_{crit}) + h·(1-F(t_{crit})),

using E_{PEP}(t) <= e_{max} for t <= t_{crit} and E_{PEP}(t) < h for t > t_{crit} (Theorem S3.1(iv)). □

Worked example: Log-normal access delay (median 72h, GSD 2.0): F(24h) ~ 0.02. With h=0.05 and e_{max}=0.95: E_bar$_{PEP}$ <= 0.02x0.95 + 0.98x0.05 = 0.068. Expected population PEP efficacy below 7%.

S7. PARAMETER TABLES

S7.1 ODE system parameters (Perelson 1996 anchored)

Symbol	Description	Value	Units	Source
d (delta)	Infected cell death rate	0.7	day^-1 (t1/2 ~1.6d)	Perelson 1996, Table 1
c	Viral clearance rate	23	day^-1 (t1/2=0.24d~6h)	Perelson 1996, Table 1
tau	Viral generation time	2.6	days (~62h)	Perelson 1996, Table 2
Eclipse phase	Min time to integration competence	0.9	days (~22h)	Perelson 1996, Table 2
beta	Infection rate constant	2.4x10^-5	mL·virion^-1·day^-1	Standard model
p	Virion production per infected cell	10^3	virions/cell/day	Standard model

alpha	Stable integration rate	10^-3	day^-1	Calibrated

PMID 8599114. Parameters c and d directly from Table 1; eclipse phase and generation time from Table 2. **CRITICAL: Reference log10 VL=4.5 for PWID is from NHBS/NHAS surveillance, NOT from Perelson 1996 (whose patients had mean log10 VL ~5.3).**

S7.2 Drug efficacy parameters

Parameter	Value	State Z(t)	Justification
e_max (pre-seeding)	0.95	Z(t)=0	Near-complete suppression of initial RT; consistent with tenofovir/FTC NHP protection data
e_mid (seeded/pre-int)	0.50	Z(t)=1	Partial suppression of established replicative focus; sensitivity: +/-40% shifts t_crit <6h
e_min (integrated)	0 (fixed)	Z(t)=2	ART does not eradicate provirus; set to 0 by A2 and Lemma S5.1

S7.3 Route-specific initial conditions and thresholds

Parameter	Mucosal	Parenteral	Note
V(0) virions/mL	1	10^3	3-log difference drives route compression; post-epithelial attenuation vs. direct IV delivery
T(0) CD4 cells/mL	10^6	10^6	Identical; peripheral blood
I(0) and R(0)	0	0	No infected cells or reservoir at exposure
I* (seeding threshold)	100 cells/mL	100 cells/mL	Sensitivity: [10,1000]; t_crit shifts <8h; compression ratio stable
R* (integration threshold)	10 cells/mL	10 cells/mL	Sensitivity: [1,100]; parenteral/mucosal ratio stable 2.5-4.0x

S7.4 Stochastic simulation parameters

Parameter	Value	Description
N Monte Carlo replications	10,000/route	Per route, per distribution, per timepoint
Time points (dense curves)	50 (0-120h)	Equally spaced; resolves all three regime boundaries
VL draws per timepoint	10,000	Independent draws from P(VL) per timepoint
Noise model (beta, alpha)	Log-normal CV=0.3	Multiplicative noise on infection and integration rates
CI type	95% credible interval	From stochastic efficacy distribution; not parameter bootstrap

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S8. STOCHASTIC VL ANALYSIS -- EXTENDED METHODS

S8.1 Source VL distributions

Expected PEP efficacy was computed as a Monte Carlo integral over source viral load:

$$E[E\_PEP(t)] = \text{integral } E\_PEP(t, VL) \cdot P(VL)\, dVL \qquad \text{(S8.1)}$$

Distribution	Mean log10	SD	Empirical basis
PWID untreated/unsuppressed	4.5	1.1	NHBS/NHAS surveillance; chronic untreated HIV-1 in PWID networks
PWID mixed treatment	3.2	1.5	Mixture: suppressed (~1.5) + unsuppressed (~4.5) in partially treated PWID networks

General HIV+ community	3.8	1.2	Mixed treatment uptake; heterogeneous MSM/general community networks
Acute-infection-enriched	5.0	0.9	High-transmission networks; acute viremia peak; recent-infection clustering

Table S8.1. All log10-normal. PWID untreated mean log10=4.5 is from NHBS/NHAS surveillance, NOT from Perelson 1996.

S8.2 Three-regime temporal classification

Regime	Hours (PWID untreat.)	Peak CI width	Clinical interpretation
1 -- Early window	0–22.0h	<15 pp	VL uncertainty low-impact; initiate PEP immediately regardless of source VL
2 -- Critical window	22.0–78.4h (PWID/acute); 22.0–88.2h (mixed/general)	39.7 pp at 49h	Maximal VL uncertainty; source VL testing clinically actionable; VL premium peaks at 21h
3 -- Late window	78.4–120h (PWID/acute); 88.2–120h (mixed/general)	<10 pp (collapsed)	Efficacy uniformly poor; VL irrelevant; P(futile) > 90%

Regime 1/2 boundary 22.04h is consistent across all four VL distributions (eclipse phase lower bound, Perelson 1996). Regime 2/3 boundary is distribution-dependent: 78.37h for PWID untreated and acute-infection-enriched distributions (higher VL accelerates CI collapse); 88.16h for mixed-treatment and general-community distributions. This distribution-specificity is biologically coherent: higher inoculum populations exhaust the critical window earlier.

S8.3 VL knowledge premium and non-monotonic plateau

VL knowledge premium = absolute difference in expected $E_{PEP}(t)$ between known vs. unknown source VL. Peak values: 3.76 pp at 21h (acute-enriched); 2.94 pp (PWID untreated); 1.87 pp (general community); 1.27 pp (mixed Tx). All distributions peak at 21h -- maximal VL-knowledge value at critical window onset.

The non-monotonic plateau between 18-36 hours in the PWID untreated and acute-enriched premium curves is a verified real kinetic signal (not numerical artefact). It arises from the interaction of the logistic inflection of the seeding-integration cascade with the 2-hour urgency reduction at the seeding midpoint in the parenteral model, creating a period during which the distribution of integration outcomes is maximally bimodal. This feature is preserved without smoothing in all reported outputs.

S9. CITY-STRATIFIED ANALYSIS -- EXTENDED METHODS

S9.1 Data source

City-level HIV surveillance data were extracted from AIDSVu downloadable datasets for 34 high-burden US metropolitan areas (2023 reporting year; downloaded October 2025).

S9.2 Mixture-model VL derivation

$$mu\_mix = f\_s \cdot mu\_suppressed + (1\text{-}f\_s) \cdot mu\_unsuppressed \qquad \text{(S9.1)}$$

Suppressed: log10 mean=1.5, SD=0.4. Unsuppressed: log10 mean=4.5, SD=1.1. IDU correction: f_s reduced by 0.12 pp per pp IDU prevalence, capped at 12 pp total.

S9.3 Structural delay derivation

$$Dt\_structural = b1 \cdot (LateDx\% - 10\%) + b2 \cdot max(90\% - Linkage\%, 0) + Dt\_SDOH \qquad \text{(S9.2)}$$

b1 = 1.2 h/%; b2 = 0.3 h/%. SDOH modifier 0-6h from Gini + poverty. Barrier categories: low (<8h), moderate (8–18h), high (18–30h), severe (≥30h).

S9.4 Counterfactual analysis

Parallel simulation assigning all 34 cities a uniform 24h access delay isolates structural barrier contribution from community VL. 33/34 cities: observed efficacy > counterfactual (structural delays <24h). Hartford CT sole exception (delay 24.4h; high-barrier category, 18–30h). Counterfactual range across cities: 76.2%–82.0%.

S10. FIGURE INVENTORY

Overview. Three main figures (Figs. 1-3), each up to four panels, generated by three Python scripts. All figures at 300 dpi. This inventory maps every panel to its source script, output file, and description.

Figure 1 -- Route-Dependent PEP Efficacy Decay

Source: `PEP_parenteral_perelson.py`, *function* `plot_vl_sweep()`. *Output:* `pep_parenteral_vl_sweep.png`

Panel	Title	Description
A	Efficacy Curves by Source Viral Load (Parenteral/PWID)	E_PEP(t) vs. hours post-exposure for 7 source VL levels (log10 = 3.0-6.0). Demonstrates VL as the primary window compressor. X-axis: 0-120h; Y-axis: efficacy 0-1. Vertical dashed lines at 24h, 48h, 72h. PRIMARY PANEL -- the money plot.
B	Biological Window Compression by Source VL	Effective prevention window (hours, defined as t_crit at eta=0.05) as a function of log10 VL. Monotone compression from ~74h (log10 VL 3.0) to ~16h (log10 VL 6.0). Horizontal reference lines at 72h (CDC) and 24h. Supports Corollary S5.1. Candidate: Supp. Fig. S1.
C	Probability PEP Already Too Late by Source VL and Time	P_int(t) as a function of log10 VL at t=24h, 48h, 72h. Shows >50% integration probability reached before 24h at high VL (log10>5). Candidate: Supp. Fig. S1.
D	Mucosal vs. Parenteral Efficacy (Route + VL Both Matter)	Paired efficacy curves for mucosal (solid) and parenteral (dashed) at matched VL values (log10=3.0, 4.5, 6.0). Demonstrates ~3-fold compression ratio. Validates NHP concordance table. PRIMARY PANEL.

For Science (4-panel allowed): include all four. If condensed, prioritize A + D; move B+C to Supp. Fig. S1.

Figure 2 -- Stochastic Efficacy Analysis Under Source VL Uncertainty

Source: `PEP_stochastic_perelson.py`, *function* `plot_stochastic_analysis()`. *Output:* `pep_stochastic_vl_uncertainty.png`

Panel	Title	Description
A	Mean + 95% CI Efficacy Curves by Community VL Distribution	Mean E_PEP(t) +/- 95% CI for all four VL distributions over 0-120h. Shaded CI bands. Regime shading overlaid: Regime 1 (white, 0–22.0h), Regime 2 (light gray, 22.0–78.4h for PWID/acute; 22.0–88.2h for mixed/general), Regime 3 (dark gray, >78.4h PWID/acute; >88.2h mixed/general) -- USE EXACT DISTRIBUTION-SPECIFIC BOUNDARIES from pep_stochastic_ci_width_regimes.csv. PRIMARY PANEL.
B	Community VL Distributions (What We Integrate Over)	Log10-normal density plots for all four VL distributions on common x-axis. Shows 1.8-log spread between extreme distributions. Provides visual grounding for Panel A. Candidate: Supp. Fig. S2.
C	Probability of Sub-Therapeutic PEP by Delay and Community	P(efficacy <50%) vs. hours post-exposure for all four distributions. Threshold line at P=0.50. Key result: P(<50%)=100% at 48h for ALL distributions. PRIMARY PANEL.
D	VL Knowledge Premium (Value of Rapid Source VL Testing)	VL knowledge premium (pp, absolute) vs. hours for all four distributions. Peaks at 21h for all distributions (3.76pp acute-enriched; 2.94pp PWID untreated). Non-monotonic plateau 18-36h PRESERVED with annotation. PRIMARY PANEL.

CRITICAL -- Panel D: The non-monotonic 18-36h plateau is a verified kinetic signal. Do NOT smooth. See S8.3. Annotation required.

CRITICAL -- Panel A: Regime boundaries are distribution-specific. R1/R2 = 22.04h for all distributions (from pep_stochastic_ci_width_regimes.csv). R2/R3 = 78.37h for PWID untreated and acute-enriched; 88.16h for mixed-treatment and general-community. Apply per-distribution shading in Panel A.

Figure 3 -- City-Stratified PEP Efficacy Across 34 US Metropolitan Areas

Source: `city_stratified_figures.py`. *Outputs:* `Fig_CityStratified_PEP.png` *(4-panel, PRIMARY) and* `Fig_CityComparison_Focus.png` *(3-panel, focus cities).*

Primary 4-panel figure:

Panel	Title	Description
A	City-Stratified PEP Efficacy at City-Specific Structural Delay	Ranked horizontal bar chart for 34 cities. Colors: green=low barrier (<8h), orange=moderate (8–18h), red=high (18–30h). Range: Milwaukee 98.3% to Hartford 78.8%. PRIMARY PANEL.
B	Structural Delay vs. PEP Efficacy (bubble = IDU prevalence %)	Scatter: x=structural delay, y=efficacy. Bubble size proportional to IDU prevalence. Color by region (South red; Northeast blue; West green; Midwest purple). Pearson r=-0.935 annotated. Key cities labeled. PRIMARY PANEL.
C	Joint VL Burden x Structural Delay Risk Surface	2D scatter/heatmap: x=community mean log10 VL, y=structural delay, color=efficacy. Shows structural delay (y) drives most variance; community VL (x) has limited effect (r=-0.084). San Juan labeled as dual-disadvantage city. Candidate: Supp. Fig. S4.
D	Actual vs. Counterfactual (24h) PEP Efficacy by City	Paired lollipop/dot: actual (solid) vs. 24h counterfactual (open) for 34 cities. 33/34 cities: actual > counterfactual. Hartford only exception (-0.59pp deficit). Counterfactual range: 76.2%-82.0%. PRIMARY PANEL.

Focus 3-panel figure (Hartford, San Juan, Jackson, Phoenix, Denver, Milwaukee):

Panel	Title	Description
A	Viral Suppression vs. Linkage to Care	Grouped bar: VS% (solid) and linkage% (faded) for 6 focus cities. Reference lines at 90% UNAIDS and US median. Shows dual compounding disadvantage.
B	Structural Delay Decomposition by Component	Stacked bar: late-dx contribution, linkage-gap contribution, SDOH modifier for 6 focus cities. Identifies dominant driver per city (Hartford=late-dx; San Juan=linkage-gap).
C	Expected PEP Efficacy (95% CI); Diamond = 24h counterfactual	Pointrange+CI for 6 focus cities with 24h counterfactual diamond overlay. Direct reading of uncertainty and counterfactual gap.

S10.1 Recommended Figure Assignment for Science Submission

Figure slot	Content	Source file	Panels
Main Fig. 1	Route-dependent window; within-host kinetics	pep_parenteral_vl_sweep.png	A + D (or all 4)
Main Fig. 2	Stochastic VL uncertainty; three clinical regimes	pep_stochastic_vl_uncertainty.png	A, C, D (Panel B to supplement)
Main Fig. 3	City-stratified structural determinants	Fig_CityStratified_PEP.png	A, B, D (Panel C to supplement)
Supp. Fig. S1	Window compression detail	pep_parenteral_vl_sweep.png	Panels B + C
Supp. Fig. S2	VL distribution densities	pep_stochastic_vl_uncertainty.png	Panel B
Supp. Fig. S3	Focus-city decomposition	Fig_CityComparison_Focus.png	All 3 panels
Supp. Fig. S4	Joint VL x structural delay risk surface	Fig_CityStratified_PEP.png	Panel C

All figures are generated by existing scripts in the project repository; no new figure generation is required for submission. Each main figure carries a self-contained argument: Fig.1 = the biological window exists and is route-specific; Fig.2 = VL uncertainty creates three clinically distinct regimes; Fig.3 = structural barriers consume the window at population scale.

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